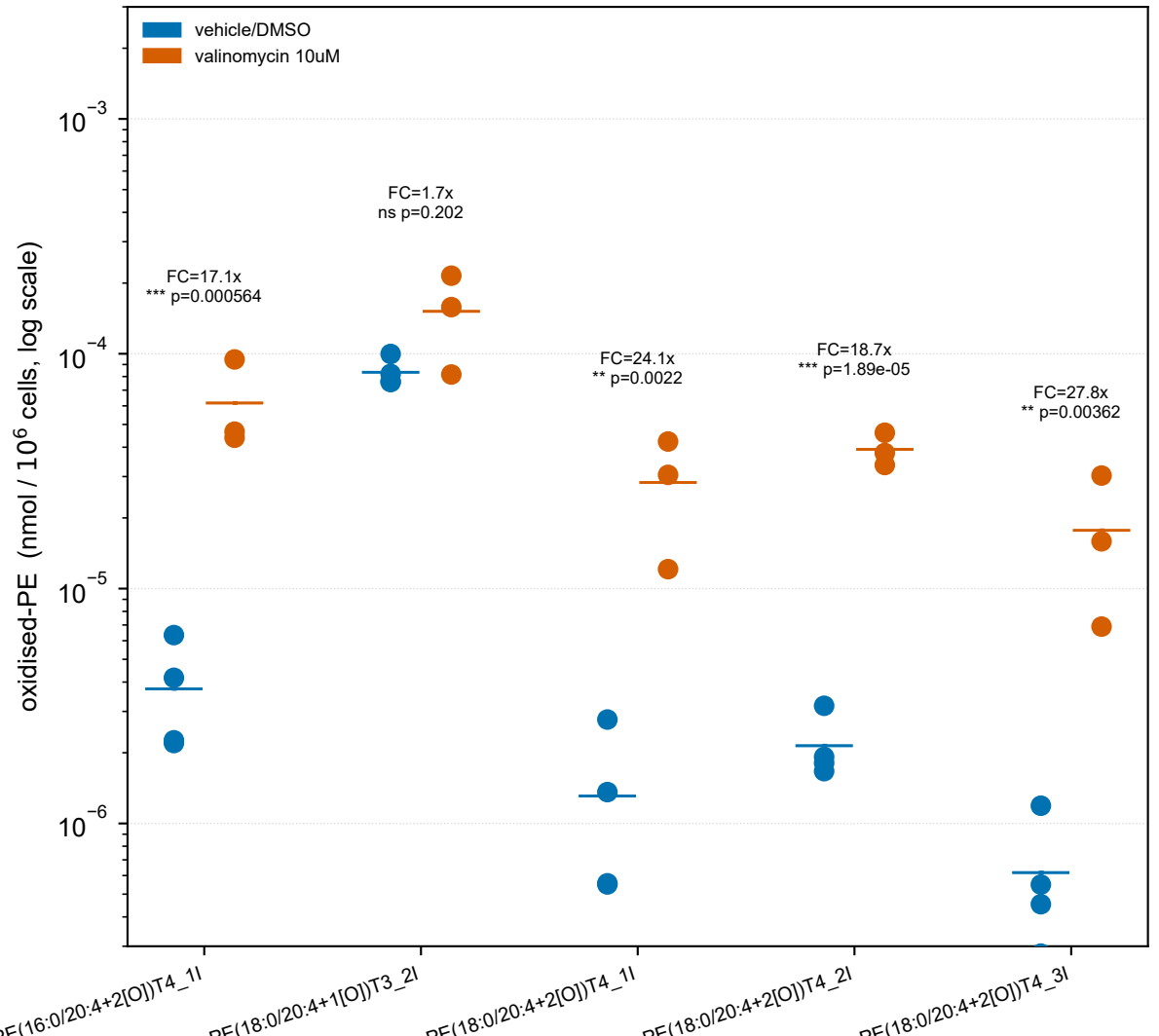
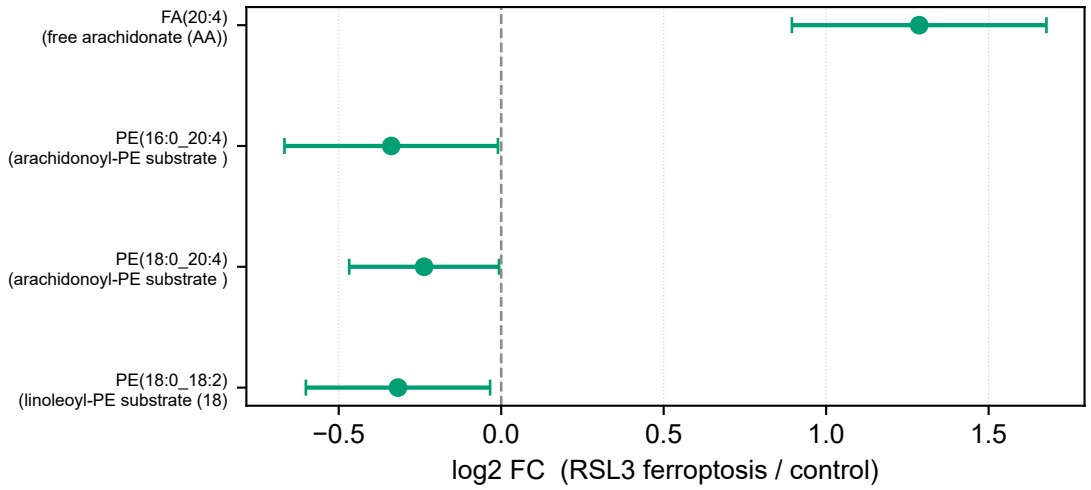
REAL lipidomics of ferroptosis EXECUTION lipids (Metabolomics Workbench ST003337 + ST004928)
ox-PE directly measured (mouse valinomycin-fibroblast + RSL3-MDSC ferroptosis models); 4-HNE / MDA / GSH not quantified in any open dataset -> reported absent, NOT proxied.

ST003337 Measured ox-PE execution lipids
valinomycin (vermilion, n=3) vs vehicle (blue, n=4)



ST004928 PUFA substrate / free-FA axis
RSL3 (GPX4i) vs control, POS-mode, green=95% CI



Requested markers: PRESENT vs ABSENT (not imputed)

ABSENT	4-HNE (4-hydroxynonenal)	not in ST003337/ST004928; not imputed
ABSENT	MDA (malondialdehyde)	not quantified; not imputed
PRESENT	ox-PE (PE-OOH / oxidised PE)	5 ox-PE species, absolute nmol/10^6
PRESENT	AA (arachidonate 20:4)	free FA(20:4) + esterified 20:4
PRESENT	LA (linoleate 18:2)	esterified only (PE 18:0/18:2)
ABSENT	GSH / GSSG	absent in all open ferroptosis data